TEXORA · LECTURE SHEET

Amines

Organic bases: classification and basicity (why aniline is a million times weaker than ethylamine), preparations from nitro reduction to Hofmann bromamide, the distinguishing tests — carbylamine, Hinsberg, nitrous acid — and diazotisation, the reaction behind the azo dyes that colour the world's textiles.

CHE-1205 · CHEMISTRY-II · TOPIC 5 · LEVEL 1 TERM 2

1. Classification and basic character

Amines are alkyl or aryl derivatives of ammonia: primary (RNH_2), secondary (R_2NH), tertiary (R_3N) by how many hydrogens are replaced. Like ammonia, the nitrogen carries a lone pair, so amines are **bases** and nucleophiles: they turn red litmus blue, form salts with acids, and smell (unforgettably) of fish. Basic strength is measured by pK_b — lower is stronger:

BASE	PK_B	WHY
Diethylamine	≈ 3.0	Alkyl groups donate (+I), stabilising the ammonium ion; in water, solvation of that ion and steric crowding pull the 2°/3° order about, giving the aqueous order 2° > 3° > 1°
Ethylamine	≈ 3.3	
Ammonia	4.75	
Aniline	≈ 9.4	The lone pair is delocalised into the benzene ring — protonating it destroys that resonance, so aniline holds its proton weakly and its lone pair tighter; ~10^6 times weaker than ethylamine

Electron-withdrawing ring substituents weaken aniline further (p-nitroaniline is feebler still); electron donors strengthen it slightly. In acid solution aniline exists as the anilinium ion $C_6H_5NH_3^+$, whose $–NH_3^+$ group is meta-directing — a fact that shapes its nitration chemistry (Section 3).

2. Preparation

- **Reduction of nitro compounds:** nitrobenzene + Sn/HCl (or Fe/HCl, or H_2/Ni) → aniline — the industrial foundation of the dye industry.

- **Ammonolysis of alkyl halides:** R–X + excess NH_3 → 1° amine, but over-alkylation contaminates it with 2°, 3° and quaternary salts unless ammonia is in large excess.
- **Reduction of nitriles and amides** ($LiAlH_4$ or H_2/Ni): R–C≡N → R–CH_2NH_2 — chain extended by one carbon.
- **Hofmann bromamide degradation:** amide + Br_2/KOH → primary amine with **one carbon fewer** (the carbonyl carbon leaves as carbonate): CH_3CONH_2 → CH_3NH_2.
- **Gabriel phthalimide synthesis:** phthalimide → N-alkylphthalimide → hydrolysis/hydrazinolysis releases a **pure primary aliphatic amine**, immune to over-alkylation.

3. Reactions and the distinguishing tests

TEST / REACTION	BEHAVIOUR
Carbylamine ($CHCl_3$ + alc. KOH, warm)	Only **primary** amines (aliphatic and aromatic) give the unbearably foul-smelling isocyanide — a definitive 1° test
Hinsberg (benzenesulphonyl chloride, then KOH)	1°: sulphonamide forms and *dissolves* in KOH (its N–H is acidic). 2°: sulphonamide forms but is *insoluble* (no N–H). 3°: no reaction — separates all three classes
Nitrous acid ($NaNO_2$/HCl)	1° aliphatic: brisk N_2 effervescence, alcohol formed. 1° aromatic at 0–5 °C: diazonium salt (no gas while cold). 2°: yellow oily nitrosamine. 3° aromatic: green p-nitroso product
Acylation	1° and 2° amines + acyl chloride/anhydride → amides (Schotten–Baumann conditions); aniline → acetanilide, whose moderated reactivity allows clean mono-nitration and bromination
Bromine water on aniline	Instant white precipitate of 2,4,6-tribromoaniline — the free –NH_2 activates the ring even more strongly than –OH does in phenol

4. Diazotisation and azo coupling — the textile connection

Aniline treated with $NaNO_2$/HCl at 0–5 °C gives **benzene diazonium chloride**, $C_6H_5N_2^+Cl^-$ — stable only in the cold, explosive when dry. Its great reaction is **coupling**: with phenol in alkali, or β-naphthol, the diazonium ion attacks the ring as an electrophile, forming an **azo compound**, Ar–N=N–Ar′. The –N=N– chromophore absorbs visible light strongly, and by choosing the two partners chemists tune the colour across the spectrum. Azo dyes are the largest single family of synthetic dyes — roughly half of all commercial textile dyes — and the diazotisation you learn here is the exact reaction run in dye houses supplying the garment factories of Bangladesh.

5. Key points

- Amines are bases by nitrogen's lone pair; aqueous basicity: 2° > 3° > 1° aliphatic > NH_3 ≫ aniline (resonance drains its lone pair).
- Pure 1° amines: Gabriel synthesis; shorter 1° amines: Hofmann bromamide; aniline: nitrobenzene reduction.
- Carbylamine = 1° only; Hinsberg separates 1°/2°/3° by sulphonamide solubility; HNO_2 behaviour differs for every class.
- Diazotisation needs 0–5 °C; coupling with phenols/naphthols gives azo dyes — the chemistry of textile colour.

6. Practice

Answer in your own words; brief model answers follow.

1. Arrange in decreasing basic strength: aniline, ammonia, ethylamine, diethylamine — and explain aniline's position in one sentence.
2. Three bottles hold ethylamine, diethylamine and triethylamine. Describe the Hinsberg procedure and the observation for each.
3. Which of these respond to the carbylamine test — methylamine, dimethylamine, aniline, trimethylamine? State the structural requirement.
4. Aniline is diazotised and the product is coupled with alkaline phenol. Name the reaction sequence, the class of coloured product, and why it matters to the textile industry.
5. Acetamide is warmed with Br_2/KOH. Name the product and the reaction, and count the carbons before and after.
6. Direct nitration of aniline with mixed acid gives largely the meta product and much tar. Explain, and state the standard remedy.

Answer key

1. Diethylamine > ethylamine > ammonia > aniline. Alkyl groups stabilise the alkylammonium ion (+I effect and solvation), while aniline's lone pair is delocalised into the ring by resonance and is therefore far less available to a proton.
2. Shake each with benzenesulphonyl chloride in KOH. Ethylamine (1°): a sulphonamide forms and dissolves in the KOH. Diethylamine (2°): a sulphonamide forms but stays insoluble. Triethylamine (3°): no sulphonamide forms at all (no N–H to substitute); the amine layer simply remains.
3. Methylamine and aniline — primary amines only. The reaction needs an $–NH_2$ group whose nitrogen can lose both hydrogens as the isocyanide carbon inserts; 2° and 3° amines lack the

required two N–H bonds.

4. Diazotisation ($NaNO_2$/HCl, 0–5 °C) followed by azo coupling; the product is an azo dye (here p-hydroxyazobenzene, orange). Azo compounds make up about half of all commercial textile dyes, so this is core industrial chemistry for dyeing and printing fabrics.
5. Methylamine — the Hofmann bromamide degradation. Carbons: acetamide has 2; methylamine has 1 — the carbonyl carbon is expelled as carbonate, shortening the chain by one.
6. In strong acid, aniline is protonated to the anilinium ion, whose $–NH_3^+$ group is electron-withdrawing and meta-directing; mixed acid also oxidises the free amine to tarry products. Remedy: protect $–NH_2$ as acetanilide (acetylation), nitrate the now ortho/para-directing, oxidation-resistant derivative, then hydrolyse the amide back to the amine.
